

10: Ecosystems

Edexcel B A-level Biology (9BI0), Topic 10. This topic is about how organisms interact with each other and with their surroundings: how ecosystems are structured and sampled, how energy passes through them and nutrients are recycled, how communities change over time, and how human activity affects them. It draws heavily on maths skills: sampling, frequency tables and histograms, mean, median and mode, standard deviation and range, an index of diversity, and the statistical tests (t-test, Spearman's rank correlation and chi-squared).

10.1 The Nature of Ecosystems

10.1 (i) What Is an Ecosystem?

Ecosystem: All the living organisms in a particular area (the community) together with the non-living (abiotic) conditions of that area, and all the interactions between them. Energy flows through an ecosystem and chemical elements cycle within it.

Ecosystems have no fixed size. A water-filled hollow in a tree, a rotting log or a rock pool can be treated as an ecosystem, and so can a pond, a lake, an oak wood or a coral reef. At the largest scale, the whole biosphere (every part of Earth that supports life) can be thought of as one ecosystem. Because the boundaries are defined by the investigator, ecosystems are never completely closed: organisms, water, mineral ions and energy move across their edges.

Several related terms are used to describe the parts of an ecosystem:

Term	Meaning
Habitat	The place where an organism lives, described by its physical and biological conditions

Population	All the individuals of one species living in the same area at the same time (and so able to interbreed)
Community	All the populations of all the species living and interacting in the same area
Niche	The role of a species in its ecosystem: what it feeds on, what feeds on it, where and when it is active, and how it affects and is affected by the biotic and abiotic conditions
Biotic factors	Living influences, such as competition, predation, grazing, disease and parasitism
Abiotic factors	Non-living influences, such as temperature, light intensity, water availability, soil pH, mineral ions, oxygen concentration and wind

10.1 (ii) Trophic Levels

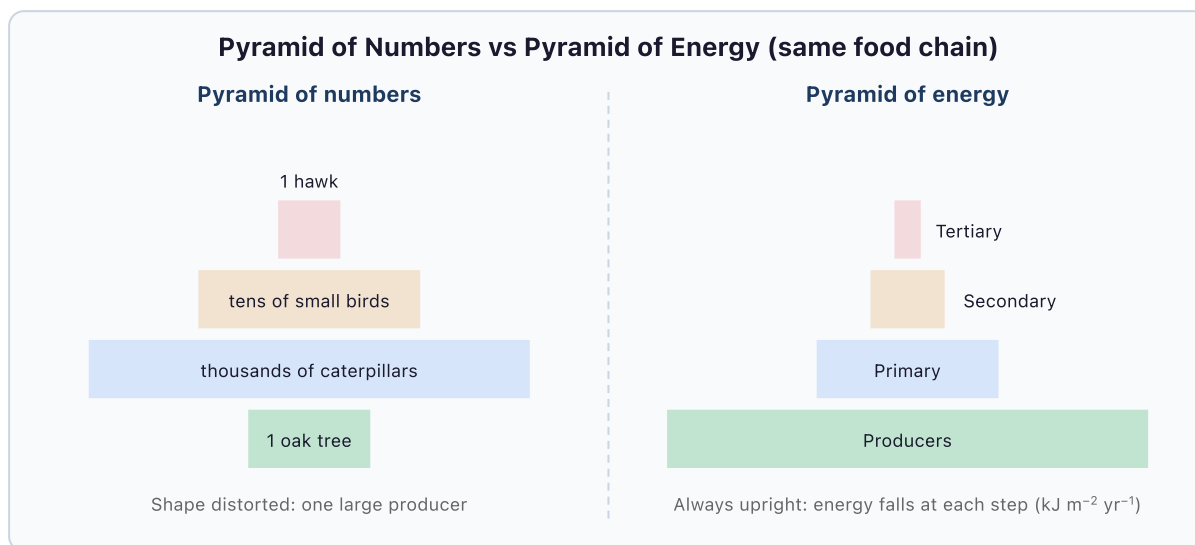
Trophic level: The feeding position that an organism occupies in a food chain, counted by the number of energy-transfer steps from the producers.

- **Producers** (trophic level 1) are autotrophs, mostly photosynthetic plants, algae and cyanobacteria. They convert light energy into chemical energy in organic molecules.
- **Primary consumers** (level 2) are herbivores that eat producers.
- **Secondary consumers** (level 3) are carnivores that eat primary consumers.
- **Tertiary consumers** (level 4) eat secondary consumers. Food chains rarely have more than four or five levels.
- **Decomposers** (bacteria and fungi) and **detritivores** (such as earthworms and woodlice) feed on dead organic matter and waste from every level. They are not usually given a single trophic level.

A **food chain** shows one feeding pathway. In real ecosystems most species eat, and are eaten by, several others, so the feeding relationships form a **food web**. An omnivore can occupy more than one trophic level at once; for example, a fox that eats both blackberries and rabbits is a primary consumer and a secondary consumer.

10.1 (iii) Ecological Pyramids

Pyramids show the relative size of each trophic level, drawn as horizontal bars with producers at the bottom. There are three kinds, and each has strengths and weaknesses.



Pyramid	What is measured	Advantages	Disadvantages
Numbers	Number of individual organisms at each level in a given area	Quick and easy to obtain by counting; no organisms need to be killed	Ignores size, so one tree counts the same as one blade of grass; can be inverted (a single oak supporting many insects) or distorted by parasites; numbers can be too large to count or draw to scale
Biomass (dry)	Dry mass of organisms at each level per unit area, e.g. g m^{-2}	Allows for size differences; dry mass removes the variable water content, which would otherwise distort the comparison	Organisms must be killed and dried to constant mass, so samples are small and may not be representative; shows the standing crop at one moment only, so seasonal changes are missed; can be inverted where producers reproduce very quickly (e.g. at times a small mass of phytoplankton supports a larger mass of zooplankton); equal masses of different tissues do not contain equal energy
Energy	Energy passing	Takes account of time and of the energy content of	Difficult and time-consuming to collect: it needs biomass measured over a whole

	through each level per unit area per unit time, e.g. $\text{kJ m}^{-2} \text{yr}^{-1}$	tissues; never inverted, because each level receives less energy than the level below; allows fair comparison of different ecosystems and of the efficiency of transfer	year plus the energy content of samples (by calorimetry); an organism may feed at more than one level, which is hard to allocate
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How Biomass and Energy Are Transferred

Biomass and the chemical energy it contains move up a food chain when one organism eats another. Only part of what is available at one level becomes biomass at the next level. Some of the producer is never eaten (roots, woody tissue); some of what is eaten is not digested and absorbed but is egested in faeces; and of the material that is absorbed, much is broken down in respiration to release energy for movement, active transport and, in mammals and birds, keeping the body warm. Only the energy that ends up in new tissue (growth and reproduction) is available to the next trophic level. The energy in dead organisms, faeces and urine is passed to decomposers instead.

Exam Tip: A common error is to say that energy is “used up” or “lost in respiration” without saying where it goes. Energy is not destroyed: it is transferred to the surroundings as **heat**, which cannot be used by other organisms. Also, never say that energy is “produced” by respiration; respiration **releases** energy from organic molecules.

10.1 (iv) Techniques for Measuring Abundance and Distribution

Abundance is how many individuals of a species there are in an area. **Distribution** is where the species is found within the area. Because it is rarely possible to count every organism, ecologists take **samples** and use them to estimate values for the whole area.

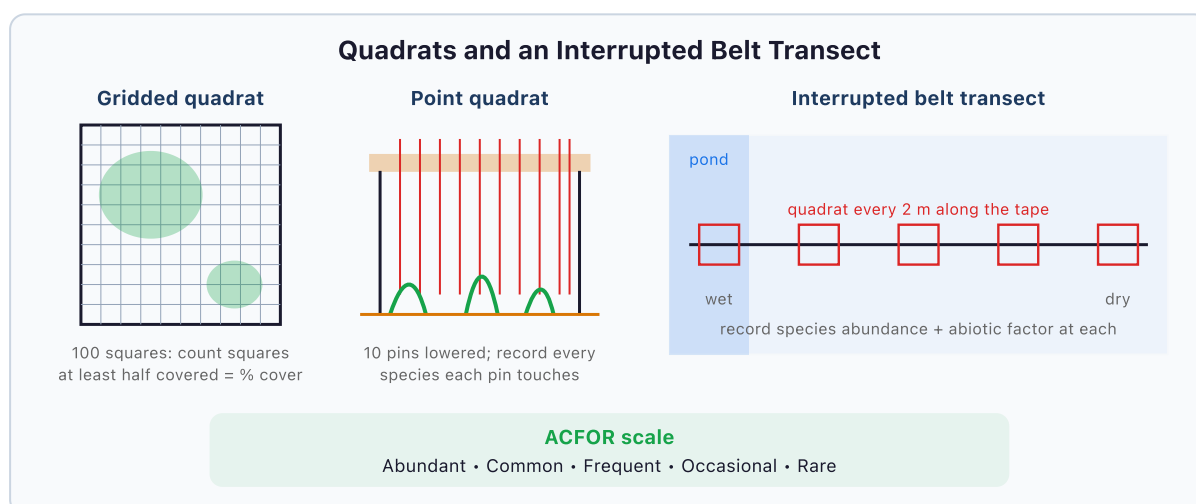
Types of quadrat

- **Frame quadrat:** a square frame (commonly $0.25 \text{ m} \times 0.25 \text{ m}$, $0.5 \text{ m} \times 0.5 \text{ m}$ or $1 \text{ m} \times 1 \text{ m}$) placed on the ground; the organisms inside are counted or their cover estimated.
- **Gridded quadrat:** a frame quadrat divided by strings into 100 small squares. Counting the squares that are at least half filled by a species gives an estimate of its percentage cover, and recording presence or absence in each square gives a local frequency.

- **Point quadrat:** a frame holding a row of pins (often 10) that are lowered to the ground one at a time. Every plant that each pin touches is recorded. It is the most objective way to measure cover of low, dense vegetation such as grassland, and it can record plants in layers.

Transects are used when conditions change across an area (an environmental gradient), for example from the sea up a shore, from a pond edge on to dry land, or from under a tree canopy into open ground.

- **Line transect:** a tape is laid out and every organism touching it (or touching it at set intervals) is recorded. It is quick but gives only presence and position, not abundance.
- **Belt transect:** quadrats are placed along the tape, either end to end (continuous) or at regular intervals (interrupted). This records abundance as well as position, so changes in abundance along the gradient can be shown as a kite diagram.



Measures of abundance

Measure	How it is recorded	Best used for	Limitations
Individual counts (density)	Count the individuals in each quadrat; density = number per m ²	Organisms that are easy to tell apart, e.g. daisies, dandelions, limpets	Impossible for plants that grow as mats or spread by runners (grasses, mosses), where individuals cannot be distinguished
Percentage cover	Estimate the proportion of the quadrat area covered	Plants and sessile animals where	Subjective when estimated by eye; layers of overlapping plants

	by each species, often using a gridded or point quadrat	individuals cannot be separated	can make totals exceed 100%
Frequency	Record the percentage of quadrats (or small squares) in which a species is present	A quick survey of many species	Gives no information about how many individuals, or how large they are
ACFOR scale	Give each species a category: Abundant, Common, Frequent, Occasional or Rare	Rapid surveys of large areas or many species	Semi-quantitative and subjective; different people may assign different categories, so results are hard to compare or analyse statistically

10.1 (v) Choosing an Appropriate Technique

The method must suit both the habitat and the organism. Useful questions to ask are:

- **Is the area uniform, or is there a gradient?** In an area that looks fairly uniform (a field, a lawn), use **random sampling**: lay two tapes at right angles as axes and place quadrats at coordinates from a random number generator. This avoids bias from the investigator choosing “interesting” spots. Where an abiotic factor changes across the site, use **systematic sampling** along a transect so that changes with distance can be linked to the factor.
- **Does the organism move?** Quadrats suit plants and slow-moving or sessile animals (limpets, barnacles). Mobile animals are sampled with pitfall traps (ground invertebrates), sweep nets (insects in long grass), tree beating onto a sheet, pooters, kick sampling in streams, or light traps for moths, and their population size is estimated by **mark-release-recapture**.
- **How big are the organisms and how are they spread out?** Quadrat size should suit the organism: small quadrats for mosses and lichens, larger ones for shrubs. Clumped species need more quadrats than evenly spread ones.
- **How many samples?** Plot a running mean of the count (or number of species found) against the number of quadrats. When the running mean levels off, extra quadrats add little, so this gives a sensible minimum sample size.
- **Which measure of abundance?** Individual counts for separable organisms; percentage cover or point quadrats for grasses and mosses; ACFOR where speed matters more than precision.

Calculating Population Estimates

$$\text{Estimated population} = \text{mean number per quadrat} \times (\text{area of site} \div \text{area of one quadrat})$$

Worked example: a mean of 3.2 daisies per 0.25 m² quadrat in a 100 m² lawn gives $3.2 \times (100 \div 0.25) = 3.2 \times 400 = \mathbf{1280 \text{ daisies}}$.

$$\text{Lincoln index: } N = (M \times n) \div m$$

Here M is the number caught and marked in the first sample, n is the total caught in the second sample and m is the number of marked animals in the second sample. *Worked example:* 40 woodlice are marked and released; a day later 50 are caught, of which 8 are marked. $N = (40 \times 50) \div 8 = \mathbf{250 \text{ woodlice}}$. The estimate is only valid if marking does not harm the animals or make them easier for predators to see, marked animals mix fully with the rest of the population, and there is no significant birth, death, immigration or emigration between the two samples.

Core Practical 15 (CP15): Compare how different sampling methods (for example quadrat size, number of quadrats, random or systematic placement, mark-release-recapture) change the estimated size of a population, handling organisms safely and ethically. For the full method, analysis and sources of error, see [Core Practicals: CP15](#).

Measuring Diversity

The maths skills for this topic include using a formula to calculate an index of diversity, which combines the number of species with how evenly the individuals are spread among them. The formula used in this course is:

$$D = N(N - 1) \div \sum n(n - 1)$$

N is the total number of organisms of all species and n is the number of each individual species. The higher D is, the more diverse the habitat. *Worked example:* a sample contains four species with 25, 12, 8 and 5 individuals, so $N = 50$. $N(N - 1) = 50 \times 49 = 2450$. $\sum n(n - 1) = (25 \times 24) + (12 \times 11) + (8 \times 7) + (5 \times 4) = 600 + 132 + 56 + 20 = 808$. $D = 2450 \div 808 = \mathbf{3.03}$.

10.1 (vi) Statistical Tests

Ecological data vary a lot, so statistical tests are used to decide whether a pattern is likely to be real or could have arisen by chance. Every test follows the same logic:

1. State a **null hypothesis**: there is no difference between the means, or no correlation between the two variables.
2. Calculate the test statistic.
3. Compare it with the **critical value** from a table at the 5% ($p = 0.05$) significance level, using the correct degrees of freedom or number of pairs.
4. If the calculated value is **greater than or equal to** the critical value, reject the null hypothesis: the probability that the result is due to chance is 5% or less, so the difference or correlation is significant. If it is smaller, accept the null hypothesis.

Test	Use it when...	Example question
Student's t-test	Comparing the means of two sets of continuous data that are roughly normally distributed	Is the mean leaf length of ivy different in shade and in full sun?
Spearman's rank correlation coefficient	Testing for a correlation between two variables measured on the same samples (at least 5 pairs, ideally 10 or more); the data need not be normally distributed	Is the percentage cover of a plant correlated with light intensity along a transect?
Chi-squared test	Comparing observed frequencies (counts in categories) with expected frequencies	Is a species found in quadrats on north-facing and south-facing slopes as often as expected by chance?

The t-Test

$$t = (\bar{x}_1 - \bar{x}_2) \div \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$$

$\bar{x}$ is the mean, s the standard deviation and n the number of values in each sample. Degrees of freedom = $n_1 + n_2 - 2$. The sign of t just shows which mean is larger, so compare the size of t with the critical value.

Worked example (illustrative data): the lengths of 10 ivy leaves from a sunny wall have a mean of 5.8 cm ($s = 0.41$ cm) and 10 leaves from deep shade have a mean of 7.6 cm ($s = 0.54$ cm).

- $s_1^2/n_1 = 0.1681 \div 10 = 0.01681$ and $s_2^2/n_2 = 0.2916 \div 10 = 0.02916$
- $\sqrt{(0.01681 + 0.02916)} = \sqrt{0.04597} = 0.2144$
- $t = (7.6 - 5.8) \div 0.2144 = \mathbf{8.40}$, with $10 + 10 - 2 = 18$ degrees of freedom
- The critical value at $p = 0.05$ for 18 degrees of freedom is 2.10. Since $8.40 > 2.10$, reject the null hypothesis: the leaves in shade are significantly longer. (A larger leaf area helps a plant absorb enough light in low light intensity.)

Spearman's Rank Correlation Coefficient

$$r_s = 1 - (6\sum d^2) \div (n(n^2 - 1))$$

Rank each variable separately (1 for the largest value, or the smallest, but the same way for both), find the difference in rank d for each pair, square it and add up the squares. n is the number of pairs. r_s lies between +1 (perfect positive correlation) and -1 (perfect negative correlation), with 0 meaning no correlation. Tied values are each given the mean of the ranks they would have occupied.

Worked example (illustrative data): eight quadrats along a transect running from open ground into a wood.

Quadrat	Light intensity / klux	Rank	% cover of grass	Rank	d	d ²
1	12.5	1	64	2	-1	1
2	9.8	2	70	1	1	1
3	8.1	3	48	3	0	0
4	6.4	4	41	5	-1	1
5	5.2	5	45	4	1	1
6	3.9	6	22	6	0	0

7	2.7	7	15	7	0	0
8	1.5	8	8	8	0	0

$\Sigma d^2 = 4$, so $r_s = 1 - (6 \times 4) \div (8 \times 63) = 1 - 24/504 = \mathbf{0.952}$. The critical value for $n = 8$ at $p = 0.05$ is 0.738. Since $0.952 > 0.738$, there is a significant positive correlation between light intensity and percentage cover.

Exam Tip: A significant correlation does **not** show that one variable causes the other. Other abiotic factors, such as soil moisture or leaf litter, also change along the same transect. When a question asks you to **justify** the choice of test, name the type of data (two means of continuous data for a t-test; paired measurements looking for a relationship for Spearman's; frequencies in categories for chi-squared). Always write the conclusion in context, referring to the calculated value, the critical value and the probability level.

Other Data-Handling Skills

- **Mean, median and mode:** the mean is the total divided by the number of values; the median is the middle value when the data are in order; the mode is the most frequent value. The median is less affected by an extreme outlier than the mean.
- **Range and standard deviation:** the range (largest minus smallest) depends only on the two extreme values, so one outlier can make it misleading. Standard deviation uses every value and measures the typical spread around the mean, so it is usually the more useful measure of dispersion. If the mean $\pm$ one standard deviation of two samples overlap a lot, the difference between them is unlikely to be significant.
- **Frequency tables, bar charts and histograms:** use bar charts for categories (such as species), with gaps between bars; use histograms for continuous data grouped into classes (such as leaf length), with no gaps.

10.2 Energy Transfer Through Ecosystems

10.2 (i) Gross and Net Primary Productivity

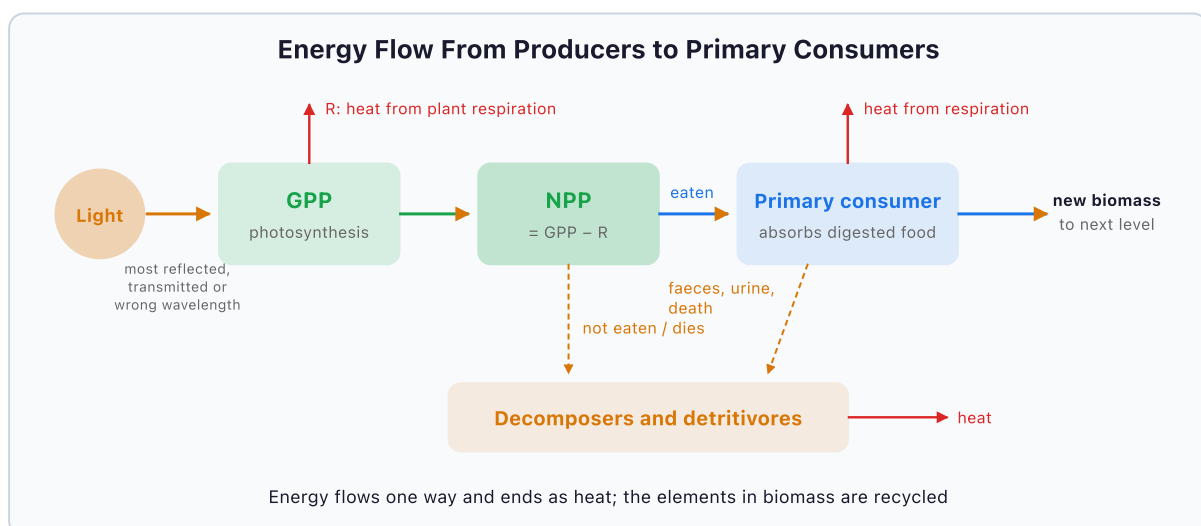
Gross primary productivity (GPP): The rate at which producers convert light energy into chemical energy in organic molecules by photosynthesis, per unit area per unit time (e.g. $\text{kJ m}^{-2} \text{yr}^{-1}$).

Net primary productivity (NPP): The rate at which energy is stored in new producer biomass, after the energy released by the producers' own respiration has been subtracted. It is the energy available to primary consumers and decomposers.

$$\text{NPP} = \text{GPP} - R \quad (R = \text{energy released by respiration of the producers})$$

Only a small percentage of the sunlight reaching a plant ends up in GPP. Much of it is reflected, passes straight through leaves, falls on non-photosynthetic parts or on bare ground, or is of a wavelength that chlorophyll does not absorb well (green light). Photosynthesis is also restricted by limiting factors such as carbon dioxide concentration, temperature and water supply. Productivity is highest in warm, wet, well-lit ecosystems such as tropical rainforests and some wetlands, and lowest in deserts and the open ocean, where water or mineral ions are limiting.

The same idea applies to consumers. The energy a consumer takes in as food is partly lost in faeces; the rest is absorbed. Of the absorbed energy, some is released by respiration and lost as heat, and some is lost in urine; what remains becomes new consumer biomass and is available to the next level.



10.2 (ii) Efficiency of Energy Transfer

$$\text{Efficiency of transfer (\%)} = (\text{energy transferred to the higher trophic level} \div \text{energy available at the lower trophic level}) \times 100$$

Make sure both values are in the same units and cover the same area and time. *Worked example:* in a grassland, $\text{GPP} = 32\,000 \text{ kJ m}^{-2} \text{ yr}^{-1}$ and producer respiration releases $12\,000 \text{ kJ m}^{-2} \text{ yr}^{-1}$.

- $\text{NPP} = 32\,000 - 12\,000 = \mathbf{20\,000 \text{ kJ m}^{-2} \text{ yr}^{-1}}$
- If primary consumers gain $1800 \text{ kJ m}^{-2} \text{ yr}^{-1}$ as new biomass, the efficiency from producers (NPP) to primary consumers = $(1800 \div 20\,000) \times 100 = \mathbf{9\%}$
- If secondary consumers gain $144 \text{ kJ m}^{-2} \text{ yr}^{-1}$, the efficiency from primary to secondary consumers = $(144 \div 1800) \times 100 = \mathbf{8\%}$

Transfer efficiencies between consumer levels are commonly in the region of 10%, though they vary widely. The energy that is not transferred is accounted for by:

- **Parts not eaten:** roots, bark and woody stems, bones, fur and shells are often left.
- **Parts not digested:** cellulose and lignin in plants, and hair, feathers and bone in prey, pass out as faeces (egestion). Herbivores lose a larger proportion this way than carnivores, because plant material is harder to digest.
- **Excretion:** energy in nitrogenous waste such as urea in urine.
- **Respiration:** the largest loss. Energy released for movement, active transport and synthesis is eventually transferred to the surroundings as heat.
- **Body temperature:** endotherms (mammals and birds) respire rapidly to keep warm, so less of their food becomes new biomass. Ectotherms, such as fish and insects, generally convert a larger proportion of their food into biomass.

These losses at every step explain why food chains are short, why there are far fewer top predators than herbivores, and why pyramids of energy always narrow towards the top.

Exam Tip: When you are asked to **calculate** an efficiency, show the working, give units where needed and check the answer is sensible (a transfer efficiency above 100% means the numbers are upside down). When asked to **explain** why the efficiency is low, give specific losses (respiration releasing heat, egestion of indigestible material, parts not eaten, excretion) rather than just "energy is lost".

10.2 (iii) Microorganisms and the Recycling of Nutrients

Energy passes through an ecosystem only once, but chemical elements such as carbon, nitrogen and phosphorus are used again and again. Without microorganisms, these elements would stay locked up in dead organisms and waste, and producers would soon run short of mineral ions.

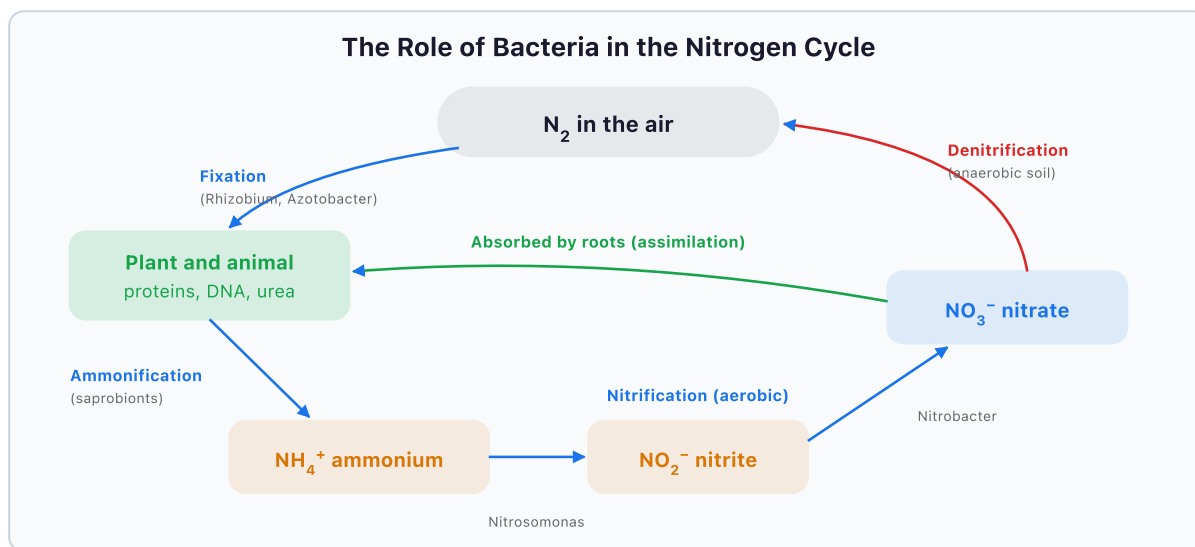
Decomposition. Saprobionts (saprotrophic bacteria and fungi) secrete enzymes such as proteases, cellulases and lipases on to dead material and faeces. The enzymes digest the material outside the cells, and the soluble products are absorbed and respired. This releases carbon dioxide into the air and mineral ions (such as ammonium and phosphate) into the soil. Detritivores such as earthworms speed this up by breaking dead material into smaller pieces, giving decomposers a larger surface area to act on. Decomposition is fastest in warm, moist, well-aerated conditions, since these suit the enzymes and aerobic respiration of the microorganisms.

The carbon cycle. Carbon dioxide is taken from the air by photosynthesis and passed along food chains in organic compounds. It returns to the air through respiration by plants, animals and, very importantly, decomposers. Where decomposition is slowed (waterlogged, acidic or cold conditions), dead matter builds up as peat, and over very long periods may form coal, oil or gas.

The nitrogen cycle. Although about 78% of the air is nitrogen gas, most organisms cannot use it directly. Bacteria carry out almost every step of the cycle:

Process	Microorganisms	What happens
Nitrogen fixation	<i>Rhizobium</i> in the root nodules of legumes (peas, beans, clover); free-living <i>Azotobacter</i> in soil	Nitrogen gas is reduced to ammonium ions using the enzyme nitrogenase. <i>Rhizobium</i> gives the plant nitrogen compounds and receives carbohydrates in return (mutualism).
Ammonification	Saprobiontic bacteria and fungi	Proteins, nucleic acids and urea in dead organisms and waste are broken down, releasing ammonium ions.
Nitrification	<i>Nitrosomonas</i> , then <i>Nitrobacter</i>	Ammonium ions are oxidised to nitrite ions and then to nitrate ions. These bacteria need oxygen, so well-aerated soils favour nitrification.

Assimilation	(Plants)	Roots absorb nitrate by active transport and use it to make amino acids, proteins and nucleic acids, which pass to consumers.
Denitrification	Denitrifying bacteria (e.g. <i>Pseudomonas</i>)	In waterlogged, anaerobic soils, nitrate is converted back to nitrogen gas, lowering soil fertility.



Other microorganisms also help. Mycorrhizal fungi grow in association with plant roots and greatly increase the surface area for absorbing water and mineral ions such as phosphate, receiving sugars from the plant in return.

Exam Tip: Do not mix up nitrification and nitrogen fixation. **Fixation** turns nitrogen gas into ammonium; **nitrification** turns ammonium into nitrate. Also remember that decomposers release nutrients as **ions** (ammonium, phosphate) and carbon as **carbon dioxide**, not “nutrients” in general. Ploughing and draining fields improve aeration, which favours nitrifying bacteria and reduces denitrification.

10.3 Changes in Ecosystems

10.3 (i) Colonisation, Succession and Climax Communities

Colonisation: The arrival and establishment of species in an area where they were not previously present.

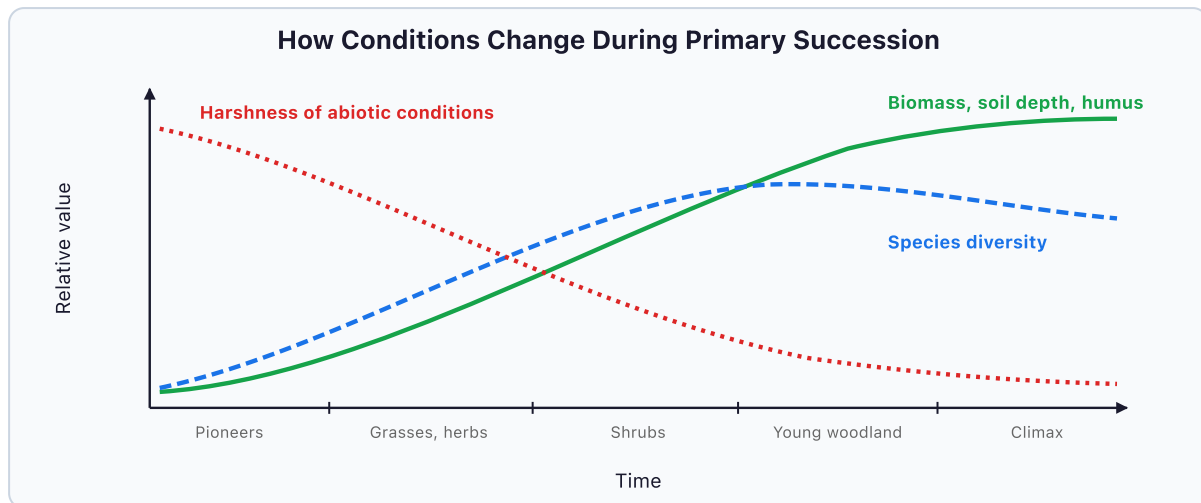
Succession: The directional change in the species composition of a community over time, as each group of species alters the conditions so that other species can establish and outcompete them.

Climax community: A relatively stable community at the end of succession, whose make-up stays roughly the same over time unless conditions change.

Primary succession starts on a surface that has never had soil or a community, such as bare rock after a glacier retreats, cooled lava, newly formed sand dunes or a new volcanic island. **Secondary succession** starts where a community has been removed but soil remains, for example after a fire, after a field is abandoned or after trees are felled. Secondary succession is faster, because soil, seeds, roots and spores are already present.

The stages of primary succession

1. **Pioneer species colonise.** They are adapted to harsh conditions: lichens and mosses on rock, or marram grass on sand dunes. Typical adaptations include tolerance of drought and extreme temperatures, the ability to photosynthesise and to obtain mineral ions where soil is lacking (some fix nitrogen), rapid growth and reproduction, and large numbers of wind-dispersed spores or seeds.
2. **Soil begins to form.** Pioneers help break the rock down by weathering, and when they die, decomposers add humus. This thin soil holds a little water and some mineral ions. Marram grass stabilises moving sand with its extensive roots and underground stems.
3. **New species arrive.** Small plants such as grasses and ferns can now grow. They shade and outcompete the pioneers. Their death adds more organic matter, so the soil gets deeper, holds more water and contains more nutrients.
4. **Larger plants follow.** Shrubs, then fast-growing trees such as birch, then slower-growing trees. Each seral stage changes the abiotic conditions (more shade, deeper soil, more shelter), allowing species that are better competitors under the new conditions to replace it.
5. **A climax community is reached.** Across most of lowland Britain this would be mixed deciduous woodland, often dominated by oak.



During succession the soil becomes deeper and richer in humus and mineral ions, it retains more water, total biomass increases, food webs become more complex and conditions become less extreme. Species diversity usually rises through the middle stages; in some climax communities it falls slightly, because a few dominant species (such as large trees) outcompete others.

Types of climax community

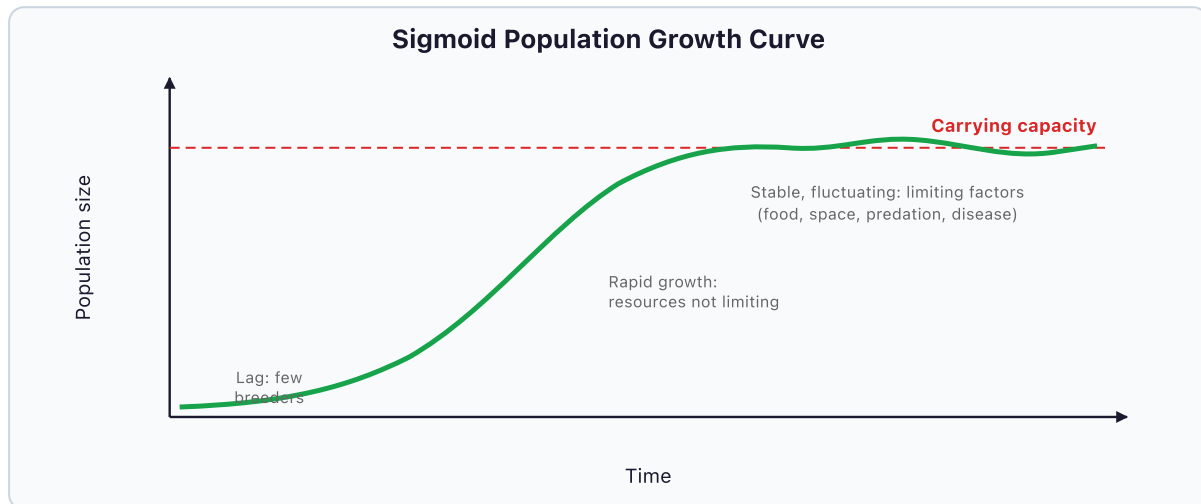
- **Climatic climax:** the climax is determined mainly by climate (temperature and rainfall). Examples include tropical rainforest in hot, wet regions, temperate deciduous woodland in much of Britain, coniferous (boreal) forest in cold northern regions, and tundra where it is too cold for trees.
- **Plagioclimax (deflected climax):** human activity stops succession before the natural climax is reached, so a stable but different community is maintained. Grazing by sheep or rabbits keeps chalk downland as grassland by eating tree seedlings; regular burning maintains heather moorland; mowing keeps lawns and hay meadows. If the activity stops, succession resumes towards woodland.

Succession is used in conservation. Some valued habitats, such as heathland, hay meadows and chalk grassland, are plagioclimaxes that disappear if management stops, so conservation involves continuing to graze, cut, burn or coppice them.

Exam Tip: Pioneer species are not simply “the first plants”: to gain the marks, explain how they **change the abiotic conditions** (weathering rock, adding humus, stabilising sand, fixing nitrogen) so that other species can survive, and how those species then **outcompete** them. When asked to **describe** succession, give the sequence; when asked to **explain** it, give the reasons for each change.

10.3 (ii) Biotic and Abiotic Factors Affecting Population Size

When a species colonises a new area, its population often follows a **sigmoid (S-shaped) growth curve**. At first growth is slow, since there are few breeding individuals. It then speeds up (exponential phase) while resources are plentiful. Finally growth slows and the population levels off around the **carrying capacity**, the maximum population the environment can support, as limiting factors take effect.



Abiotic factors affect how well organisms survive, grow and reproduce:

- **Temperature:** affects enzyme activity, so it controls the rate of growth and development, especially in ectotherms and plants. Endotherms in cold conditions must use more energy to keep warm, leaving less for growth and reproduction.
- **Light intensity:** limits the rate of photosynthesis and so the growth of producers; day length also triggers flowering and breeding in many species.
- **Water availability and humidity:** affect plant growth and transpiration, and the survival of animals that lose water easily, such as woodlice and amphibians.
- **Soil pH and mineral ions:** affect which plants can grow and how quickly (for example, nitrate for proteins and magnesium for chlorophyll).
- **Oxygen concentration:** important in water and in waterlogged soils.
- **Wind, exposure and salinity:** important on coasts and mountains.

Biotic factors result from other organisms:

- **Intraspecific competition** (between members of the same species) for food, water, light, space, nest sites or mates. As a population grows, competition increases, so fewer individuals survive and breed. This is a major reason why populations level off at the carrying capacity.

- **Interspecific competition** (between different species) for the same resources. If two species have very similar niches, one may outcompete and exclude the other. In a classic laboratory study, G. F. Gause grew two species of *Paramecium* (*P. aurelia* and *P. caudatum*) together, and *P. aurelia* eventually replaced *P. caudatum*. The introduced grey squirrel replacing the native red squirrel across much of Britain is often given as a field example.
- **Predation:** predator and prey numbers often rise and fall in linked cycles. When prey are plentiful, more predators survive and breed; the growing predator population eats more prey, so prey numbers fall; with less food, predator numbers then fall, allowing the prey to recover. The peaks in predator numbers lag behind those of the prey. Long-term fur-trapping records of lynx and snowshoe hare in Canada are a well-known example, although food supply for the hares also plays a part.
- **Disease and parasitism:** spread more easily in dense populations and can reduce survival and reproduction.
- **Food supply** and the effects of grazing on plant populations.

Factors whose effect increases as population density rises (competition, predation, disease) are called **density-dependent**. Factors whose effect does not depend on density, such as a severe frost, flood or drought, are **density-independent**, and are usually abiotic.

Core Practical 16 (CP16): Measure how a single abiotic factor (such as light intensity or soil moisture) affects where one species grows, or a feature of its structure such as leaf size, with safe and ethical fieldwork. For the full method (transect or comparison of two sites), analysis using Spearman's rank or the t-test, and sources of error, see [Core Practicals: CP16](#).

Exam Tip: When you **explain** a change in population size, link the factor to a process: for example, "low temperature reduces enzyme activity, so the rate of photosynthesis falls and less biomass is produced", not just "it is colder". In predator–prey graphs, **describe** the pattern using the data (peaks, lag, approximate values), then explain it using food supply and predation.

10.4 Human Effects on Ecosystems

10.4 (i) Interpreting Data on Human Influences

Exam questions often give graphs or tables showing human effects on ecosystems and ask you to describe trends, calculate changes and evaluate conclusions. Two main examples are named in this topic: climate change and the depletion of biological resources.

Climate change

- **Carbon dioxide concentration:** continuous measurements at the Mauna Loa observatory in Hawaii began in 1958. They show a steady long-term rise, with a small annual wave caused by the seasonal growth of vegetation in the Northern Hemisphere (CO₂ falls in summer when photosynthesis is high and rises in winter). Air trapped in ice cores shows levels of about 280 ppm before the Industrial Revolution; current values are above 420 ppm.
- **Temperature records:** global average surface temperature has risen by more than 1 °C since the late 1800s. Other evidence includes ice-core data (which show that CO₂ and temperature have varied together over hundreds of thousands of years), tree rings (dendrochronology) and pollen preserved in peat bogs, which show how vegetation, and so climate, has changed.
- **Causes:** burning fossil fuels and deforestation increase atmospheric CO₂; livestock, rice paddies and landfill release methane. These gases absorb infrared radiation re-emitted from Earth's surface, enhancing the greenhouse effect.
- **Effects on organisms:** changes in species distribution (many are moving towards the poles or to higher altitudes); changes in the timing of seasonal events such as flowering, leaf burst, egg laying and migration, which can create mismatches between a predator and its food; faster development in ectotherms as enzyme activity increases; changes in the sex ratio of reptiles whose sex is set by incubation temperature; coral bleaching in warmer seas; and ocean acidification as more CO₂ dissolves in seawater.

Depletion of biological resources

- **Overfishing:** fish are caught faster than the population can replace them by reproduction. Modern fleets with sonar and large nets can remove most of the breeding adults, and catching young fish before they mature reduces future recruitment. The Atlantic cod fishery off Newfoundland collapsed in the early 1990s, and a moratorium on fishing was declared in 1992; stocks were very slow to recover. North Sea cod stocks also declined sharply in the late 20th century.
- **Deforestation:** clearing forest for timber, farmland and cattle ranching removes habitats, reduces biodiversity, reduces CO₂ uptake and releases stored carbon when the trees are burned or decay.

- **Overexploitation of other species:** hunting and collecting for food, medicine, the pet trade and ornaments (for example, ivory from elephants and horn from rhinos).

When interpreting data:

- Describe the overall trend with figures from the data, then any fluctuations or anomalies.
- Calculate percentage change where helpful: $(\text{new} - \text{original}) \div \text{original} \times 100$.
- Check the axes: a y-axis that does not start at zero can make a small change look dramatic.
- Remember that **correlation is not causation**. Two variables rising together (such as CO₂ and temperature) supports a causal link only when there is also a plausible mechanism and other explanations have been ruled out.
- Take care with **extrapolation** beyond the data. Predictions of future climate come from models, which depend on assumptions about future emissions, so they are given as ranges.
- Consider how the data were collected: sample size, the area and period covered, and whether the source is reliable.

10.4 (ii) CITES and Global Biodiversity

CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora): An international agreement between governments that controls trade across borders in wild animals and plants, and products made from them, so that trade does not threaten their survival. It was agreed in 1973 and came into force in 1975, and more than 180 countries are now parties to it.

Species are listed in three appendices according to how much protection they need:

Appendix	Species included	Control of trade
I	Species threatened with extinction (e.g. tigers, great apes, most sea turtles)	International commercial trade is banned; trade for purposes such as scientific research needs both export and import permits
II	Species that are not necessarily threatened now but could become so if trade is not controlled	Trade is allowed only with an export permit, granted when it will not harm the survival of the species in the wild

III	Species protected in at least one country, which has asked other parties to help control trade	Trade needs permits or certificates from the listing country
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Effects on biodiversity. Supporters point out that CITES has helped reduce the legal international trade in many threatened species. For example, the African elephant was moved to Appendix I in 1989, banning international commercial trade in ivory. It also encourages cooperation between countries, raises awareness, and helps prevent species being driven to extinction by trade alone.

Limitations. CITES controls only international **trade**; it does not protect habitats or stop hunting for local use, and habitat loss is the main threat to most species. It relies on each country to pass laws and enforce them, which is difficult where there is poverty, corruption or a lack of trained officials. Illegal trade and poaching continue wherever prices are high, and countries can enter reservations (opt out) for particular species. Because many factors affect a population at once, it is hard to separate the effect of CITES from that of other measures.

Other international agreements also aim to protect biodiversity. The **Convention on Biological Diversity**, signed at the Earth Summit in Rio de Janeiro in 1992, commits countries to conserve biodiversity, use biological resources sustainably and share the benefits of genetic resources fairly.

10.4 (iii) Sustainability: Balancing Human Needs and Conservation

Sustainability: Using resources in a way that meets present needs without reducing the ability of future generations to meet theirs; for a biological resource, harvesting no faster than it can be replaced.

Conflicts arise because people need food, jobs, fuel, land and economic growth, while conservation may require limiting those activities. Managing resources sustainably means finding a balance that both sides can accept, based on scientific evidence, and enforcing it.

Conserving fish stocks

- **Quotas:** limits on the total mass of each species that may be caught in an area each year (for example, the total allowable catches set under the EU Common Fisheries Policy and now negotiated by the UK). They are based on scientific estimates of stock size, aiming to catch no more than the population can replace.

- **Minimum mesh sizes:** larger holes in nets let young fish escape, so they can reach breeding age and replenish the stock. Minimum landing sizes work in the same way.
- **Closed seasons and closed areas:** no fishing during the breeding season or in spawning and nursery grounds. **Marine protected areas** (including no-take zones) allow populations to recover and fish to spread into surrounding waters.
- **Limiting fishing effort:** restricting the number of boats, days at sea or the type of gear, and banning damaging methods such as some forms of bottom trawling.
- **Fish farming (aquaculture):** reduces pressure on wild stocks but can cause pollution from waste, spread disease and parasites to wild fish, and often uses wild-caught fish as feed.

These measures cause conflict. Fishing communities can lose income and jobs, fish that are too small or over quota have in the past been thrown back dead (discards), and quotas set too high to protect jobs may fail to protect the stock. Fish move across national boundaries, so agreement between countries is essential, and enforcement at sea is costly.

Reducing possible causes of climate change

- Replacing fossil fuels with renewable energy (wind, solar, hydroelectric), and improving energy efficiency in homes, industry and transport.
- Using biofuels from crops, which release carbon that was recently absorbed by photosynthesis; however, growing them can compete with food crops for land and lead to deforestation.
- Reducing deforestation, replanting forests and protecting peatlands, which act as carbon sinks.
- Carbon capture and storage, and reducing methane emissions from agriculture and landfill.
- International agreements, such as the Kyoto Protocol (1997) and the Paris Agreement (2015), in which countries committed to limit warming to well below 2 °C above pre-industrial levels.

The conflict here is between economic development and cutting emissions. Industries and jobs depend on fossil fuels; developing countries argue that they should be allowed to grow their economies, as richer countries did; and the costs of action fall now while many benefits come later. Sustainable management needs cooperation between governments, businesses and individuals.

Exam Tip: Evaluate and **discuss** questions on sustainability need both sides: the benefit to conservation and the cost to people (jobs, food, income), followed by a judgement. Explain *how* a measure works biologically, for example “a larger mesh size allows immature fish to escape, so more survive to reproduce and the stock can recover”.

10.4 (iv) Validating Evidence: Journals, Peer Review and Conferences

Climate change is a subject of public and political debate, so it is important to know how scientific evidence is checked before it is accepted.

- **Scientific journals** publish the details of research: the methods, raw data or results, analysis and conclusions. Publishing the methods in full allows other scientists to repeat the work and check whether the results are reproducible.
- **Peer review:** before publication, a journal editor sends the paper to other independent experts in the same field, usually anonymously. They check that the methods are valid, the data support the conclusions, the statistics are appropriate, and the work is original and properly linked to earlier research. The paper may be accepted, returned for changes or rejected. This reduces the chance of flawed, biased or fraudulent work being published, although it cannot guarantee that a result is correct, and reviewers can occasionally be biased or miss errors.
- **Scientific conferences** let researchers present new findings, often before publication, and have them questioned by other experts. They encourage discussion, collaboration and the sharing of data between groups working on the same problem.
- **Consensus reports:** the Intergovernmental Panel on Climate Change (IPCC), set up in 1988 by the World Meteorological Organization and the United Nations Environment Programme, brings together large numbers of scientists to assess the peer-reviewed evidence and summarise it in regular reports. The drafts themselves are reviewed by experts and governments.

When judging a claim about climate change, ask whether it has been published in a peer-reviewed journal, whether it has been replicated by independent groups, how large and representative the data set is, and whether the authors or funders have a conflict of interest (for example, funding from an industry that would be affected by the conclusions). Evidence reported only in the media, on websites or by pressure groups has not been through this process and should be treated with caution. Scientific understanding can change as new evidence is found; this is a strength of the scientific method, not a weakness.

Exam Tip: Peer review checks that the work is scientifically sound; it does **not** repeat the experiment. Replication by independent scientists is what shows that results are reproducible. If asked why a conclusion is more **valid** after peer review, mention independent checking of methods, analysis and conclusions by experts in the field.

Key Terms Checklist

Make sure you can define, explain or use each of the following:

- Ecosystem, habitat, population, community, niche
- Biotic and abiotic factors
- Trophic level, producer, primary, secondary and tertiary consumer, decomposer, detritivore
- Food chain and food web
- Pyramids of numbers, dry biomass and energy, with an advantage and a disadvantage of each
- Frame, gridded and point quadrats; line and belt transects
- Individual counts (density), percentage cover, frequency and the ACFOR scale
- Random and systematic sampling; running mean
- Estimating population from quadrats; mark-release-recapture and the Lincoln index
- Index of diversity: $D = N(N - 1) \div \sum n(n - 1)$
- Null hypothesis, critical value, significance level ($p = 0.05$), degrees of freedom
- Student's t-test and Spearman's rank correlation coefficient: when to use each, and how to calculate
- Mean, median, mode, range and standard deviation
- Gross primary productivity, net primary productivity, $NPP = GPP - R$
- Efficiency of energy transfer and the reasons for energy losses
- Saprobiont, extracellular digestion, decomposition
- Nitrogen fixation, ammonification, nitrification, denitrification, assimilation
- Colonisation, pioneer species, seral stage, primary and secondary succession
- Climax community, climatic climax, plagioclimax
- Sigmoid growth curve, carrying capacity
- Intraspecific and interspecific competition, predator-prey cycles
- Density-dependent and density-independent factors
- Enhanced greenhouse effect, evidence for climate change, correlation and causation, extrapolation
- Overfishing, quotas, mesh size, marine protected areas
- CITES and its three appendices; Convention on Biological Diversity
- Sustainability and the conflict between human needs and conservation

- Scientific journals, peer review, scientific conferences, IPCC
- Core Practicals 15 and 16